

a hide-and-measure protocol



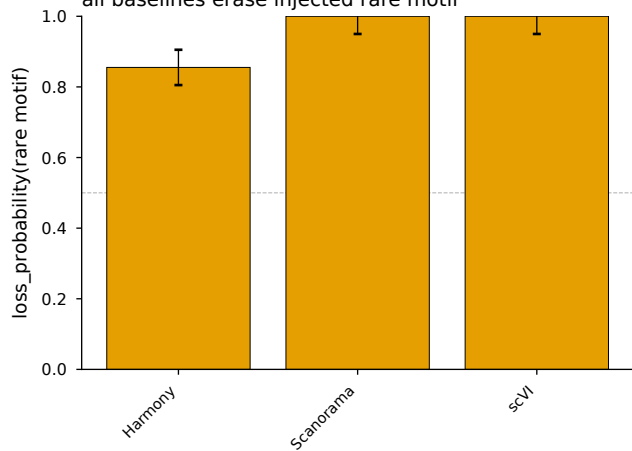
REAL never sees the held-out label; the label is used only ex-post to grade the detector.

NEGATIVE control (philosophy-gated):
inject pure batch-noise mode. REAL must NOT flag.
Failure $\Rightarrow$ trap #2 (smoothness-as-truth)
or #8 (failure-to-reject $\equiv$ success).

POSITIVE PENDANT control (philosophy-gated):
inject rare reproducible biological mode ($\leq \pi_0$, cross-batch).
REAL MUST flag under aggressive integration.
Failure $\Rightarrow$ power below the Thm 1 detectability envelope.

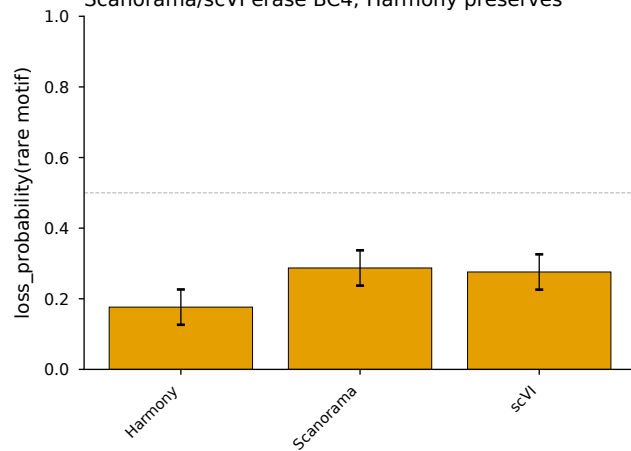
b

pancreas_synth — sensitivity (synthetic)
all baselines erase injected rare motif



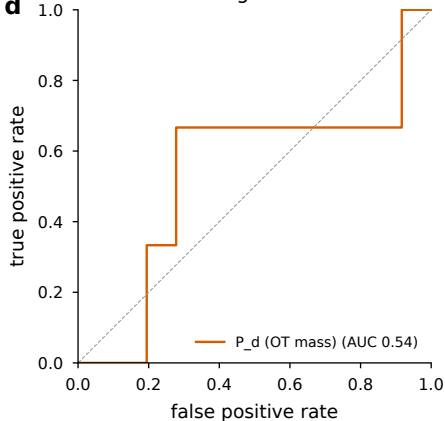
c

macaque retina BC4 — sensitivity (real data)
Scanorama/scVI erase BC4; Harmony preserves



d

ROC — detecting known loss



e

ensemble gain over single channel

